Cenomi Chatbot — Client Status Report

As of: Wednesday, 25 March 2026
Demo target: Sunday, 30 March 2026 (5 days)

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1. Where We Are on the Improvisation — What Has Been Implemented

The platform has shipped four full versions since the initial build on 13 March. Every change has been verified at 100% pass rate across the 204-query test suite (smoke + adversarial multi-turn + deep complex scenarios).

v1.0 — 13 March: Core Pipeline (Single Mall)

- **LangGraph pipeline** — 12-node directed graph covering: session load → intent interpretation → scene memory update → playbook resolution → strategy selection → context composition → retrieval decision → fact fetch → response generation → memory update → debug payload emit
- **Intent classification** — three-tier hybrid: short-circuit rule table (< 1ms) → regex rule-based classifier → GPT-4o LLM fallback. Domains: dining, shopping, entertainment, services, navigation, exploration, mall_info, general
- **Scene memory** — accumulates companions, occasion, budget, visit plan, topic lock, active shortlist across every turn; resolves terse follow-ups in context
- **Playbook engine** — scenario playbooks matched and ranked against intent + entity context; tone-mapped outputs (date night vs family day vs gift hunt produce structurally different responses)
- **Anti-hallucination guard** — post-generation validation of all LLM output against canonical mall data before delivery
- **Feedback system** — explicit (thumbs up/down) + implicit (auto-detected corrections); session tuning engine applies signals in real time during the session
- **React testing console** — full debug inspector (intent, playbook, entities, retrieval, node trace, latencies), session export
- **Mall loaded:** Al Nakheel Plaza, Buraidah (mall 28) — 94 brands, 1 cinema, 6 movies, 10 services

v1.1 — 17 March: Multi-Mall Support

- Runtime now holds a registry of mall contexts instead of a singleton
- Mall of Arabia, Jeddah (mall 13) added — full data pipeline run (canonical, semantic, playbook, tenant config)
- All pipeline nodes updated to use `mall_id`-scoped context
- Frontend mall selector updated; health endpoint returns `mall_ids` list

v1.2 — 17 March: Cross-Mall Brand Search

- New `cross_mall` intent domain — fires for queries like "Does Mall of Arabia also have Nike?" or "Which Cenomi mall has Starbucks?"
- `search_brand_across_malls()` merges canonical data across all loaded malls, sorts home mall first
- Dedicated `_build_cross_mall_response()` with structured `AT YOUR CURRENT MALL / AT OTHER CENOMI MALLS` prompt blocks
- Hallucination guard extended to validate against a merged canonical index from all loaded malls

v1.3 — 23 March: Infrastructure & Streaming

- **Redis session persistence** — sessions survive backend restarts and are shareable across workers; `RedisSessionStore` with configurable TTL; in-memory fallback when Redis is absent
- **LangGraph turn checkpointing** — every graph invocation snapshot-persisted to Redis (`BACKEND_ENABLE_CHECKPOINTER=true`); each turn is independently replayable for debugging and regression testing
- **Async quality evaluator** — LLM-as-judge evaluates every response on `intent_alignment`, `constraint_adherence`, `honesty`, and `conciseness` (0–10); runs in a background task, never blocks delivery
- **SSE token streaming** — `POST /api/chat/stream` streams response token-by-token; React UI renders live blinking cursor; session persistence and quality evaluation run post-stream without delaying the first token
- **Docker / Docker Compose** — single-command deployment; multi-stage Dockerfile under 300 MB; Redis + backend services with healthcheck; `backend/data` volume mount for persistence

v1.4 — 24 March: Semantic Retrieval + Multi-Mall Scale

- **Three-layer hybrid retrieval:**
  - **Layer 1 — Category rules:** deterministic lookup, 100% precision, zero LLM cost (~0ms)
  - **Layer 2 — Canonical name search:** substring + word-boundary match for brand-specific queries
  - **Layer 3 — Vector similarity search:** `VectorStoreService` (Chroma + `text-embedding-3-small` + Redis result cache); falls back to tag-overlap matching when Chroma is unavailable
- **`VectorStoreService`** — Chroma PersistentClient (HNSW cosine, 1536-dim); Redis-backed result cache (TTL configurable); per-mall collections with zero cross-contamination
- **Multi-mall LRU cache (three-tier resource design):**
  - Tier 1 (Python RAM) — LRU cache of active `MallContextLoader` objects (default capacity 5); eviction frees RAM
  - Tier 2 (Redis) — serialized mall context JSON with TTL; restoring from Redis is ~50–100× faster than disk
  - Tier 3 (Disk) — cold load from canonical/semantic/playbook JSON; writes back to Redis on first load
- **Experience layer** (`experience_resolver.py`) — selects response composition mode before the LLM is called: `micro_itinerary`, `curated_shortlist`, `filtered_factual_list`, `guided_plan`, `structured_overview`

- **Pronoun reference fix** — queries like “anything she would like” now correctly route to `guided_recommendation` instead of treating “she” as a brand name
- **Comprehensive test runner** (`run_all_tests.py`) — runs all three test suites against the live Docker backend; generates Markdown + JSON reports in `test-results/`

v1.4.1 — 24 March: Contextual Query Augmentation

- `build_scene_prefix(scene)` prepends `SceneMemory` signals (occasion, companions, visit_type, budget) to the vector query before embedding
- The same bare query from different visitor contexts now produces different embedding vectors:
  - Anniversary couple → embed `"anniversary partner date something for dinner"`
  - Family with kids → embed `"kids family outing something for dinner"`
  - Fresh session → embed `"something for dinner"` (unchanged — no regression risk)
- **Zero regressions: 204/204 test cases pass (100%)**

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2. What Will Be Included Before the Demo (30.03.26)

The demo is in 5 days. Based on the roadmap priorities and effort estimates documented in `docs/COMPETITIVE-ANALYSIS-AI-CORE-RETRIEVAL.md` and `docs/AI-FINDR-COMPARISON.md`, the following items are **realistic to complete before 30 March:**

Retrieval Improvements (can be done before demo)

Item	Effort	Impact	Status
BM25 entity name index + RRF fusion	1–2 days	High — closes the exact brand name miss gap vs vector-only search	Pending
Cross-encoder re-ranker (top-20 → top-5, `ms-marco-MiniLM-L-6-v2`, ~25MB, CPU-runnable)	< 1 day	Medium — improves ranking for ambiguous queries by ~15–25%	Pending
Vector fallback in factual flow (`fetch_exact_facts` falls back to vector when canonical lookup returns empty)	< 1 day	Medium — surfaces answers for service/facility queries that miss keyword lookup	Pending

Demo Preparation (should be done before demo)

Item	Notes
README update	Current README describes old 12-node single-flow architecture; live system is a 16-node dual-flow graph. Should reflect actual architecture and API routes.
Delta-ingest trigger (mtime-based, background)	Medium effort (2–3 days) — borderline for the demo window

Items NOT Realistic Before 30.03.26

Item	Reason
Arabic multilingual support	High effort — language detection, Arabic prompt variants, regional dialect handling; not achievable in 5 days
Analytics dashboard	Requires database migration + new frontend; 1+ week of work
WhatsApp integration	Requires WhatsApp Business API registration + integration work
Visual / rich responses (cards, maps, CTAs)	Requires new response schema + frontend rendering overhaul
LangSmith production tracing	Config exists but never wired into any code path

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3. How Close Are We with AI Findr

AI Quality Dimension	Parity	Status
Conversational AI core	~87%	**Cenomi leads**
Retrieval quality	~85%	**Cenomi leads** — three-layer hybrid + scene-augmented vectors; BM25 fusion in progress
Intent classification depth	**Cenomi leads**	Three-tier hybrid (rule fast-path + LLM fallback); AI Findr is single-pass LLM only
Hallucination control	**Cenomi leads**	Post-generation validation against live canonical data; AI Findr's grounding is unpublished
Real-time session adaptation	**Cenomi leads**	Mid-session behavioral tuning from feedback; AI Findr improves offline only

Where We Win

- **Multi-turn scene memory** — tracks companions, occasion, budget, and visit plan across every turn; AI Findr answers one query at a time
- **Dual-flow routing** — concierge mode for planning, factual mode for lookups; AI Findr is a single flow
- **Hallucination guard** — every LLM output validated against live mall data before delivery; AI Findr's grounding is opaque
- **Real-time session adaptation** — feedback signals (explicit and implicit) adjust behavior mid-session; AI Findr only improves offline
- **Playbook scenario engine** — "romantic dinner" and "family outing" produce structurally different responses over the same data
- **Full data ownership** — Gulf occasions, local brands, Saudi cultural context embedded natively; AI Findr is a generic platform
- **Data residency** — all visitor data stays on Cenomi infrastructure; AI Findr requires Enterprise tier for on-premises

Where AI Findr Currently Leads

- **Visual responses** — interactive maps, store cards with images, booking CTAs
- **Arabic** — auto-detect multilingual with regional tone; we have no Arabic support yet
- **Analytics** — real-time dashboard for top queries, trends, and content gaps
- **Channels** — WhatsApp, kiosk, digital signage, voice, mobile app; we are web-only today

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4. What AI Findr Doesn't Have That We Have (and Are Building)

Already Differentiated (Cenomi capabilities)

Capability	What It Means for Visitors
Scene Memory	The concierge remembers who you're with, your occasion, budget, and shopping task — across every turn of the conversation
Concierge vs Factual routing	Planning queries ("help me plan a date night") and lookup queries ("where is Zara?") are handled by separate reasoning pipelines, not a one-size-fits-all flow
Hallucination guard	Store names, hours, movies, and offers in every response are validated against live mall data before the user sees them
Real-time feedback adaptation	If a visitor corrects the bot mid-session, behavior adjusts immediately — no waiting for a next-day model update
Playbook scenario engine	Context-matched scenarios (anniversary, family day, back-to-school) change entity

Capability	What It Means for Visitors
	ranking, tone, and response structure dynamically
Scene-augmented vector search	The same query ("something for dinner") returns different results for an anniversary couple vs a family with kids — context shapes retrieval, not just the response
Cross-mall brand search	"Is Zara in any Cenomi mall?" searches across all properties with a single validated answer
Quality evaluator	Every response is automatically scored on accuracy, relevance, and honesty in the background — without slowing delivery
Full ownership	Prompts, rules, playbooks, and entity data are Cenomi's — Gulf/Saudi context is embedded natively, not approximated by a generic vendor
Data stays in-house	All visitor interactions processed on Cenomi infrastructure by default

In Progress (Roadmap)

Capability	Timeline	What It Adds
Hybrid retrieval (BM25 + vector fusion)	Pre-demo	Exact brand name queries that slip through vector search are caught by keyword index — higher recall across all query types
Re-ranking	Pre-demo	Top results for ambiguous queries reordered by true relevance — measurably better recommendations
Arabic language support	Post-demo	Full Arabic conversation with Gulf dialect handling and bilingual retrieval
Analytics dashboard	Post-demo	Business teams see top queries, trends, and content gaps without engineering involvement